

How to constrain wave attenuation and sea-ice thickness with the ICESat-2 laser altimeter?

An example for harvesting remote sensing data and derive parametrization

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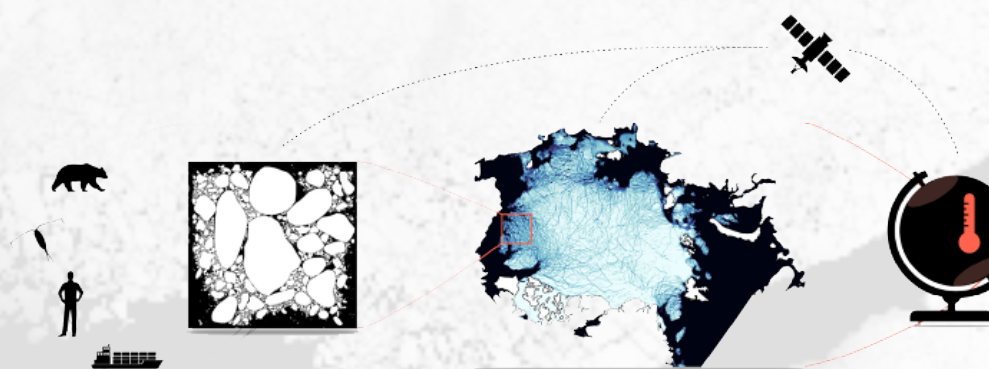
Brown University, Providence, RI, USA

collaborators

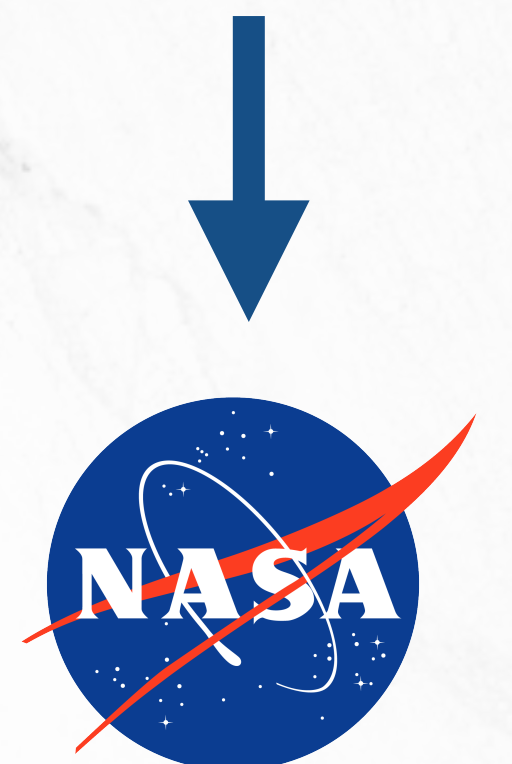
Bertrand Chaperon, Ifremer, France

CC Bitz, UW

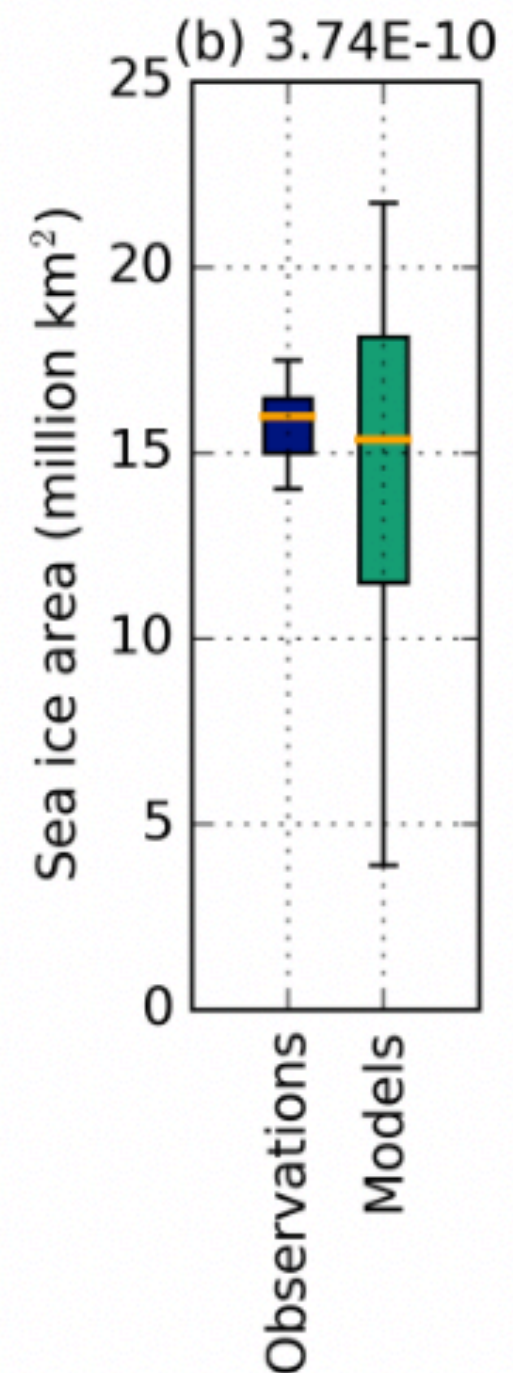
Sliderule team, NASA Goddard



 SCHMIDT FUTURES

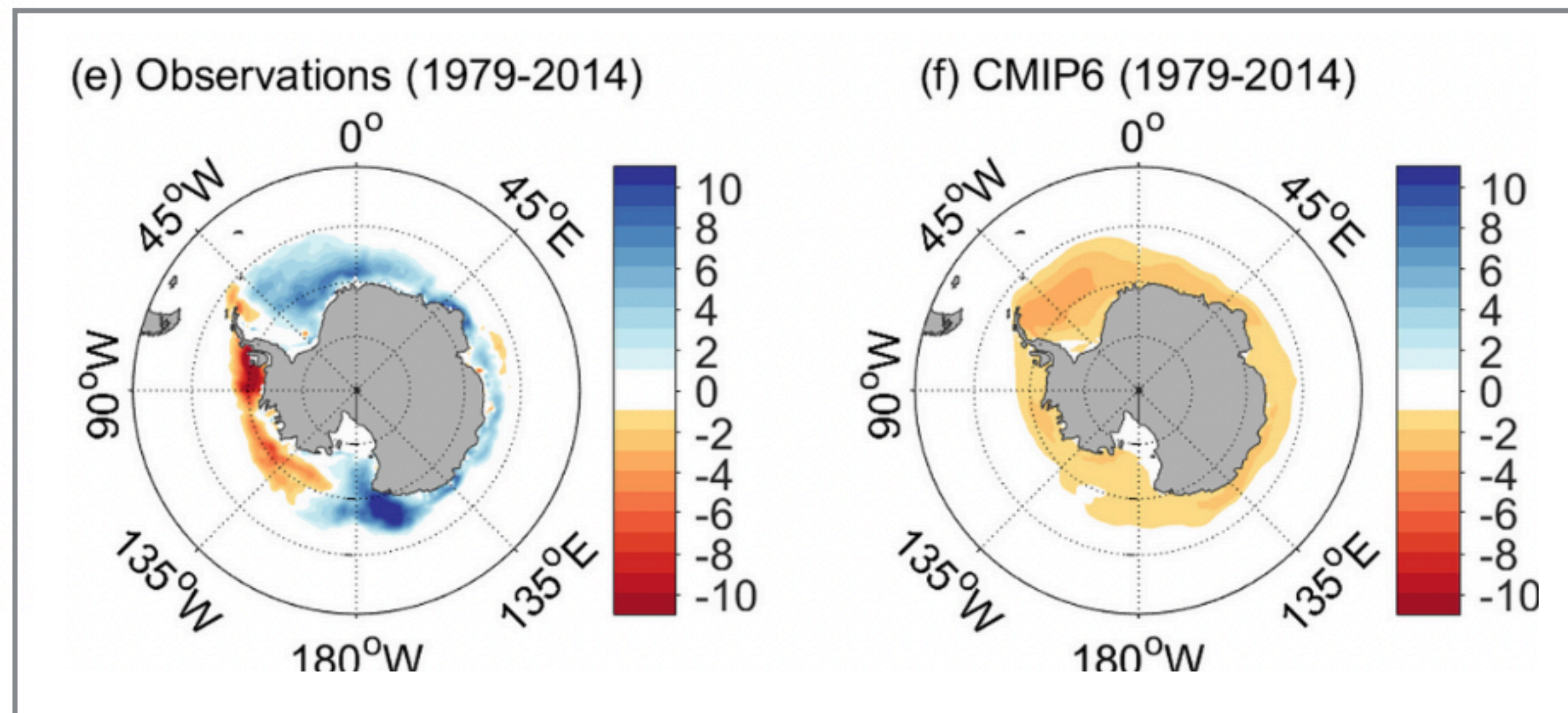


CMIP models how large(st) biases in sea ice extent and trend



Maximum sea ice area in CMIP5 models
Roach et al. 2018

Observed and CMIP trend in sea ice concentration



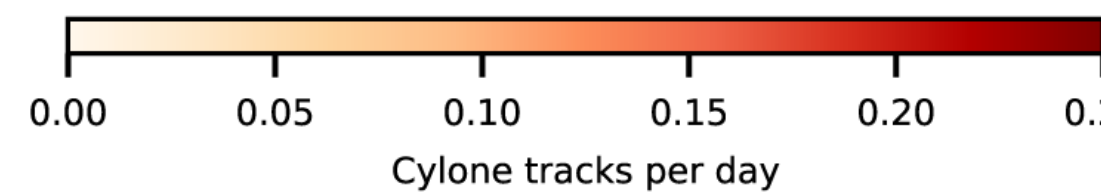
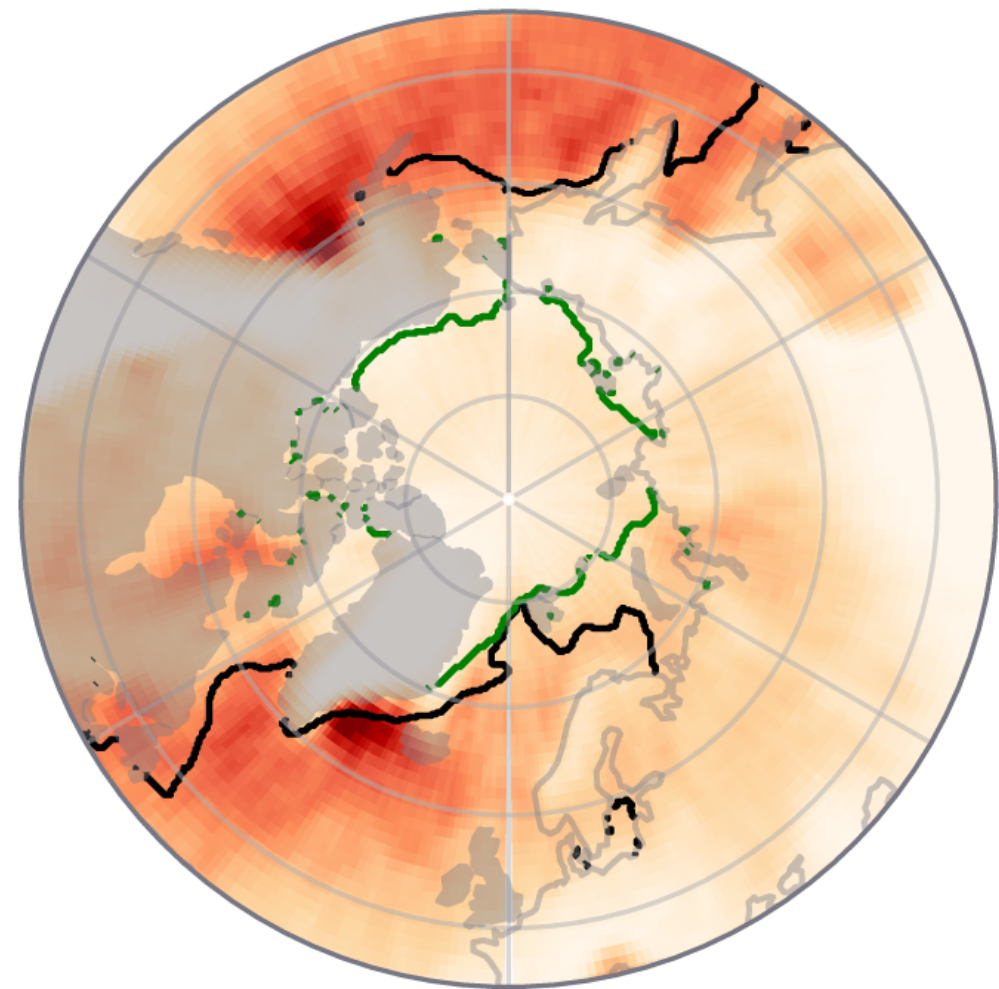
Shu et al. 2020

Sea ice models miss the impact of waves!

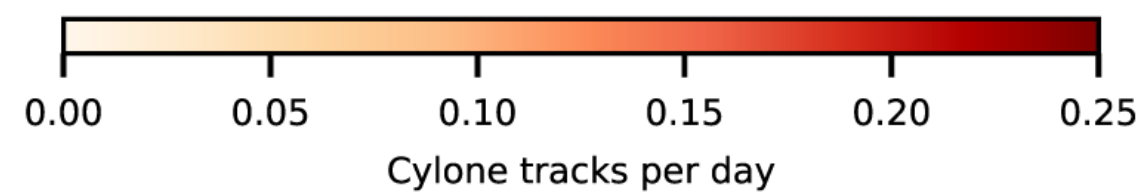
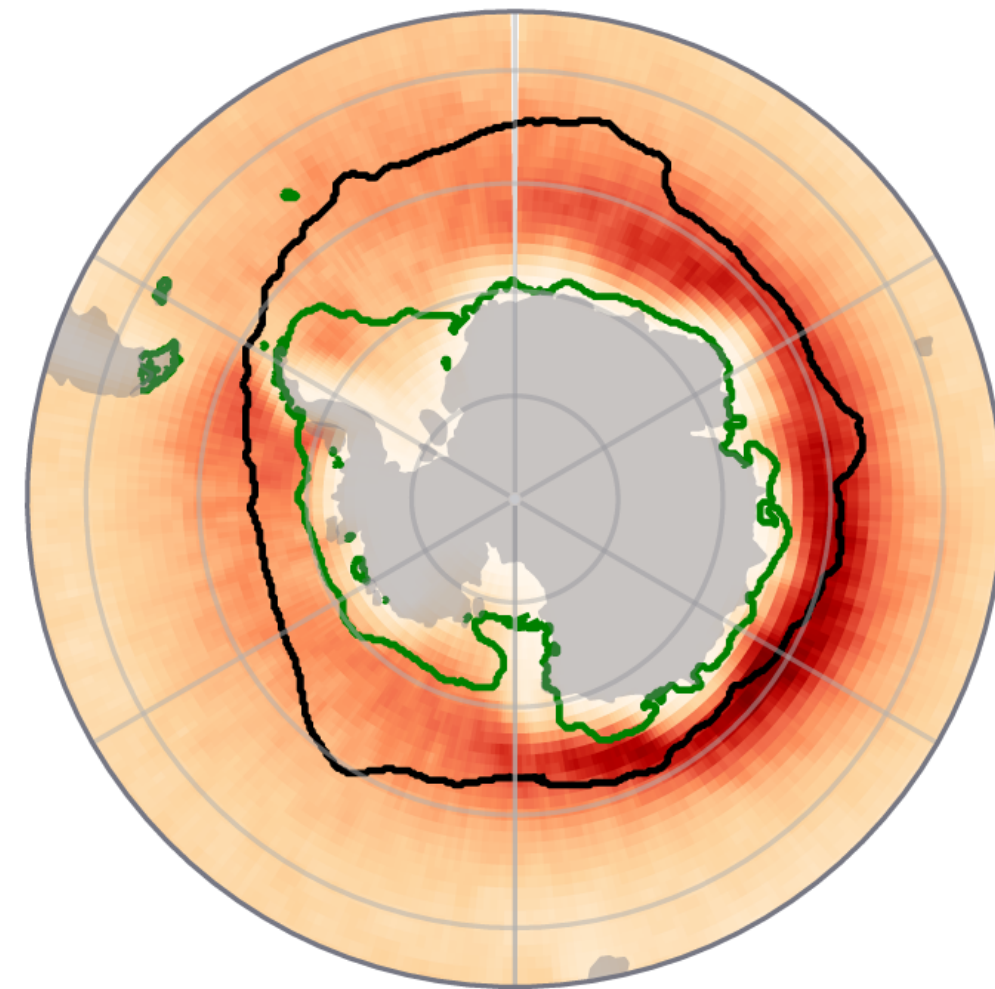
- We do not model waves in CMIP models
- We do not observe waves in sea ice sufficiently to constrain wave attenuation

Waves are in sea ice all the time!

Northern Hemisphere Cyclone density and Sea Ice Extend



Southern Hemisphere Cyclone density and Sea Ice Extend

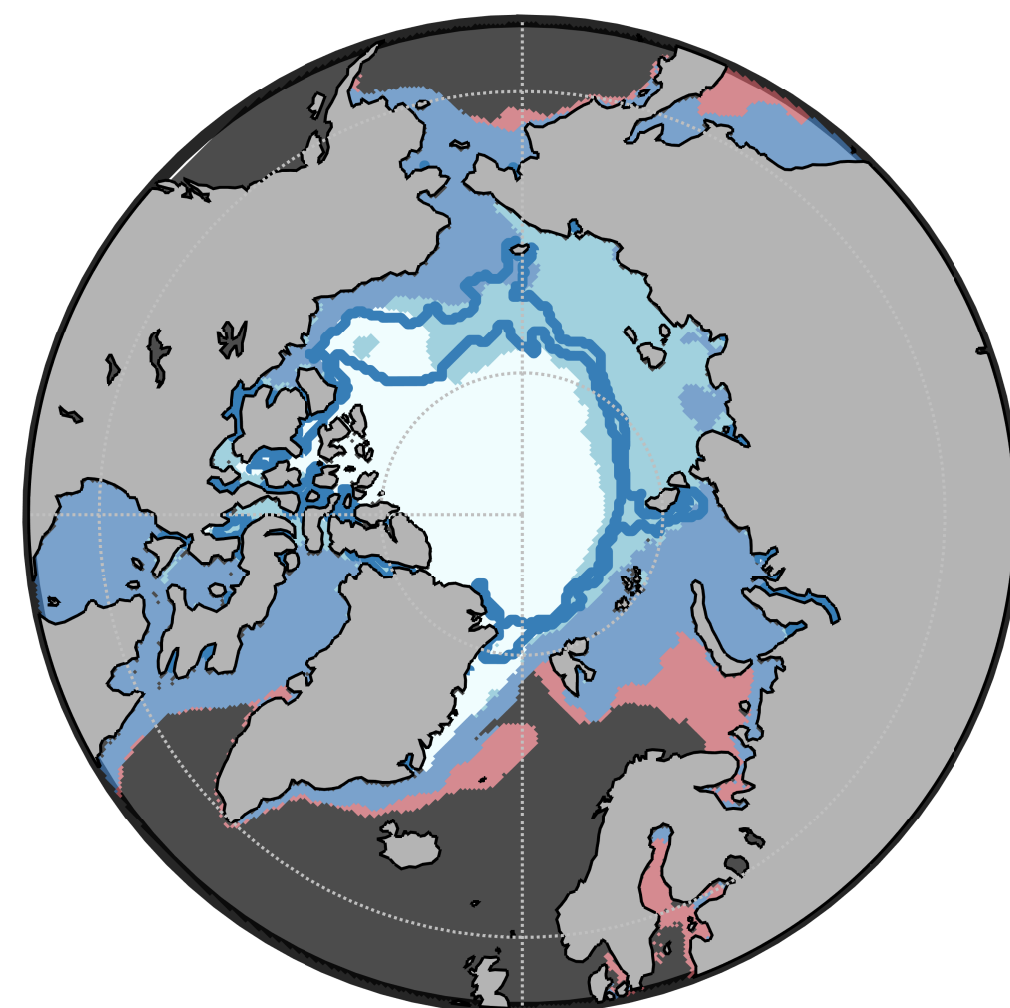


storm track density is the largest near the winter sea-ice edge

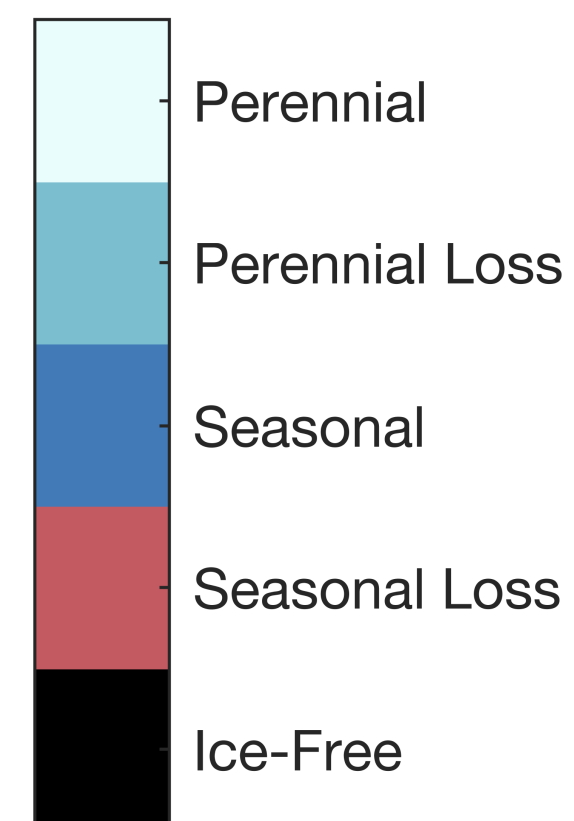
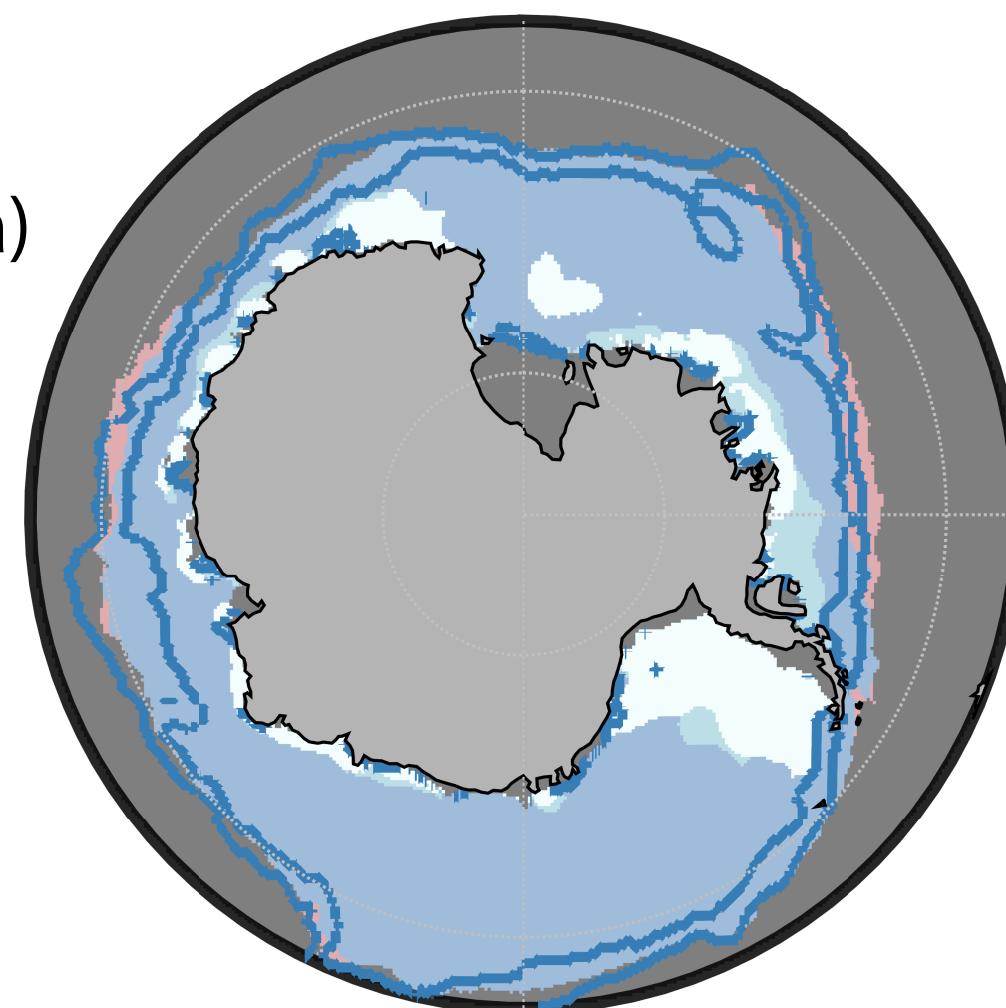
shading: storm track density (cycles per day)

Black lines: Winter maximum sea ice extend

Green lines: Summer minimum sea ice extend

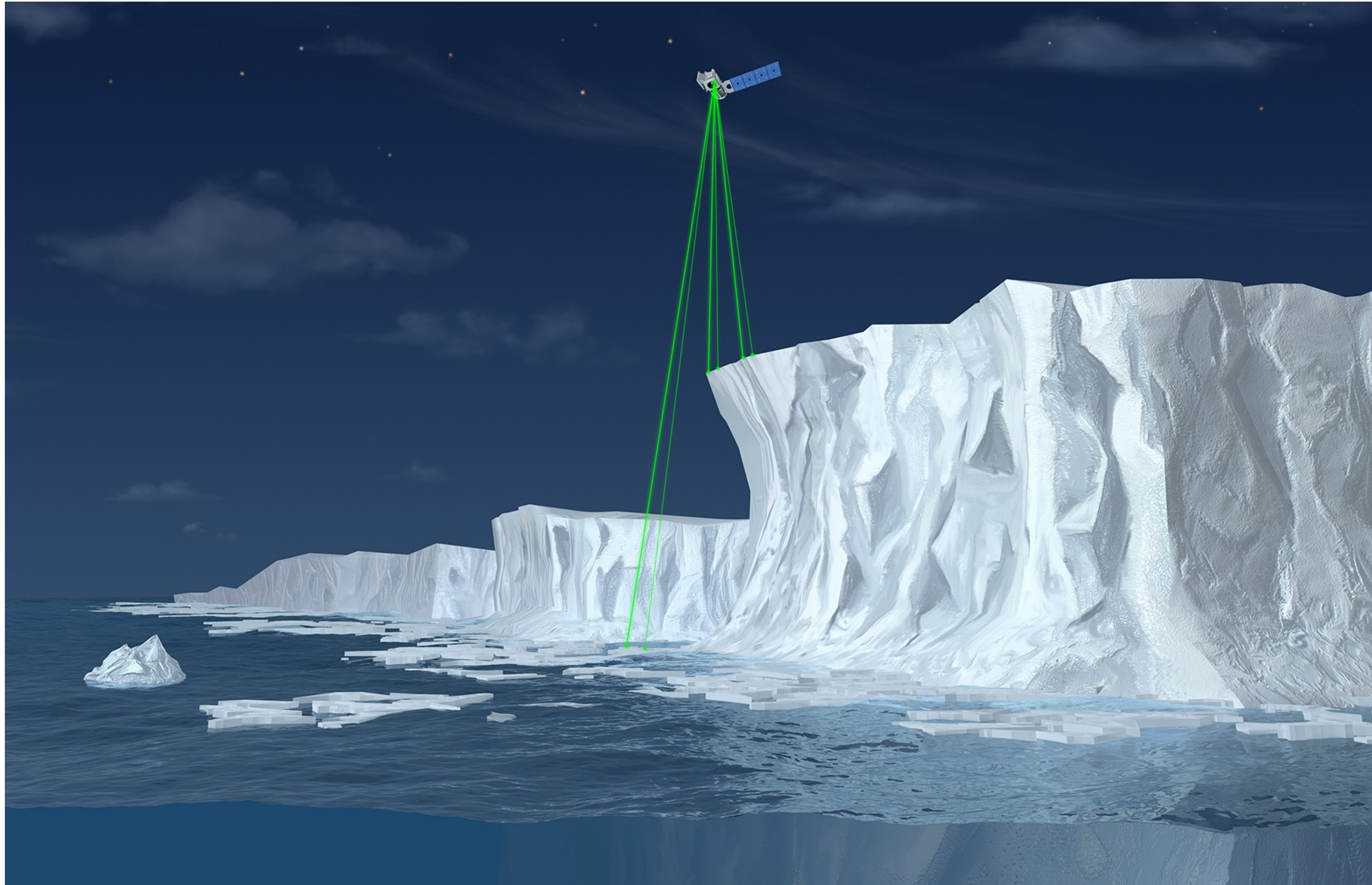


(a)



The more seasonal the sea ice gets the more prone it is to wave impact!

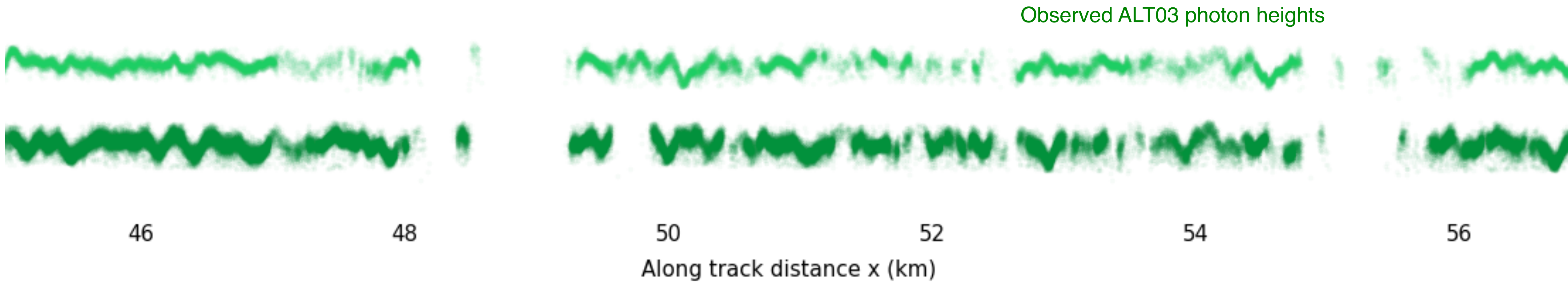
ICESat-2 is a laser altimeter to monitor ice melt in Greenland and Antarctica



I show how to turn ICESat-2 data into directional wave buoys in sea ice

How to separate the wave and ice signal in ICESat-2 data?

How to separate waves from sea ice in gappy data?



**observed
surface height variance**

=

~~**wave variance**~~

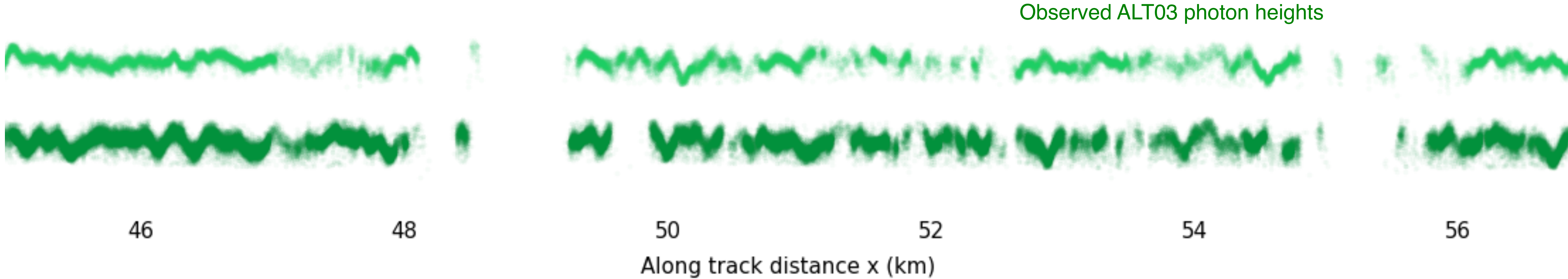
+

freeboard height

**directional surface
wave spectra**

How to separate the wave and ice signal in ICESat-2 data?

How to separate waves from sea ice in gappy data?



Proposed model: super position of linear waves for a harmonic fit

$$\mathbf{b} = \underbrace{\sum_{m=1}^M \left(a_m \cos(k_m \mathbf{x}) + b_m \sin(k_m \mathbf{x}) \right)}_{\mathbf{H} \mathbf{p}} + \mathbf{r}$$

ICESat-2 data

$\partial_x h$

wave model with

$\mathbf{p} = [a_1, a_2, \dots, a_M, b_1, \dots, b_M]^T$

with $2M$ model parameters

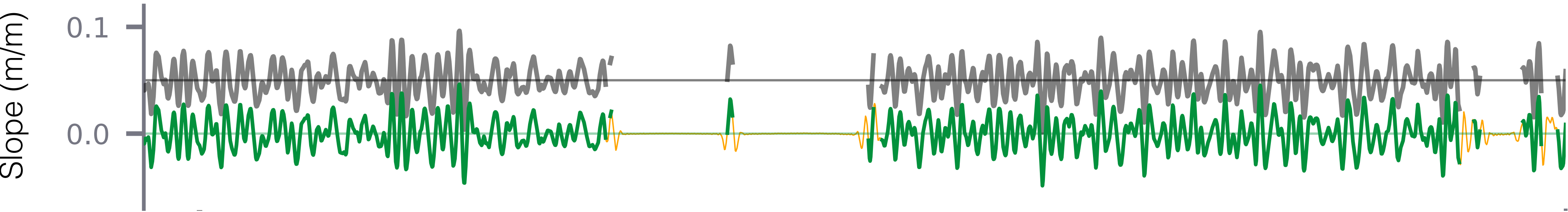
residual

sea ice signal?

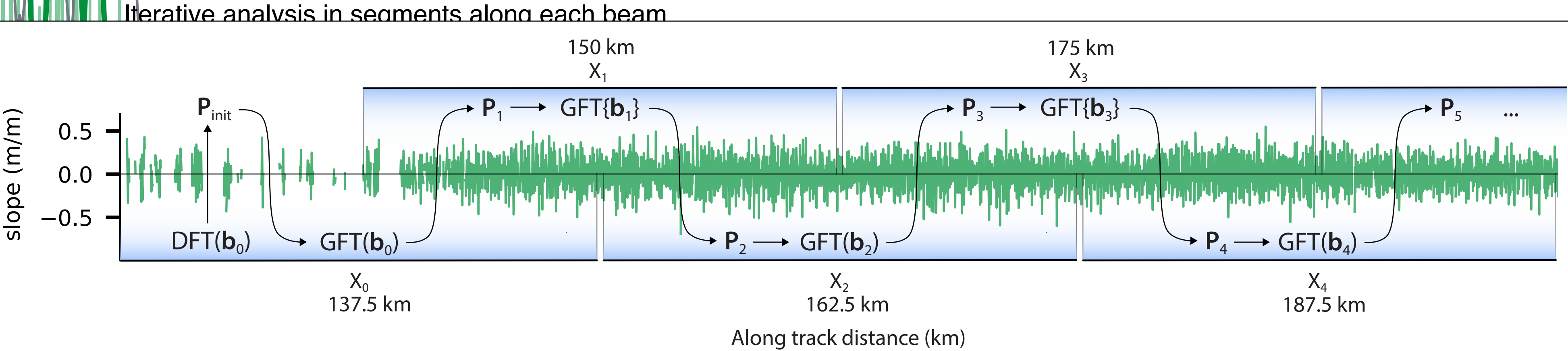
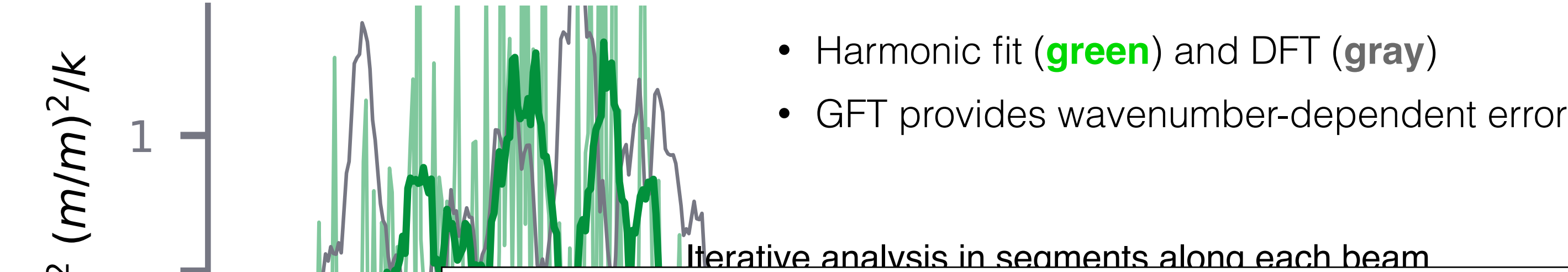
Solving for harmonic fit using linear inversion

(Wunsch 1996, Kachelein et al 2022, and others ...)

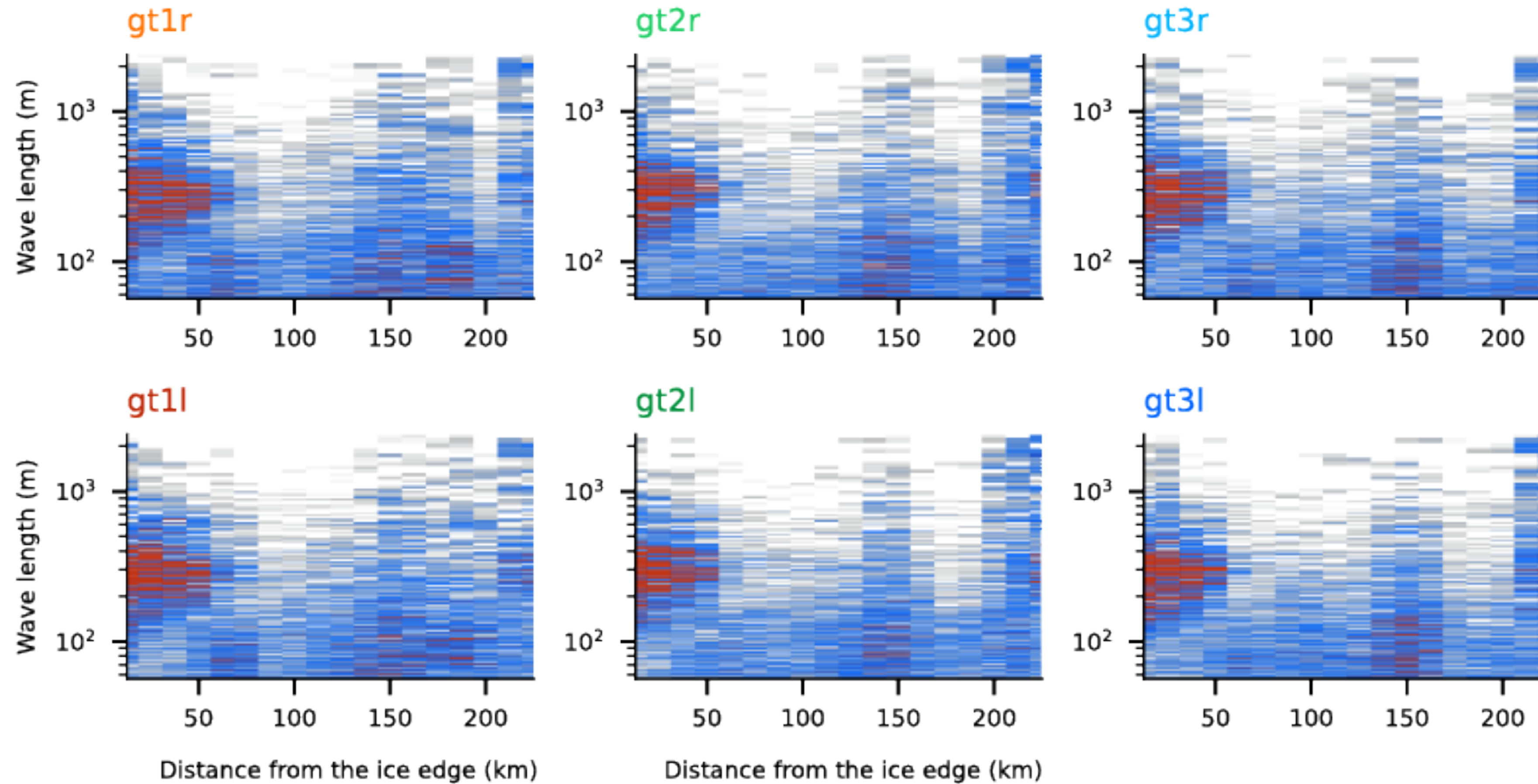
Harmonic Model vs. Data comparison



Wave spectrum from a 25-km segment

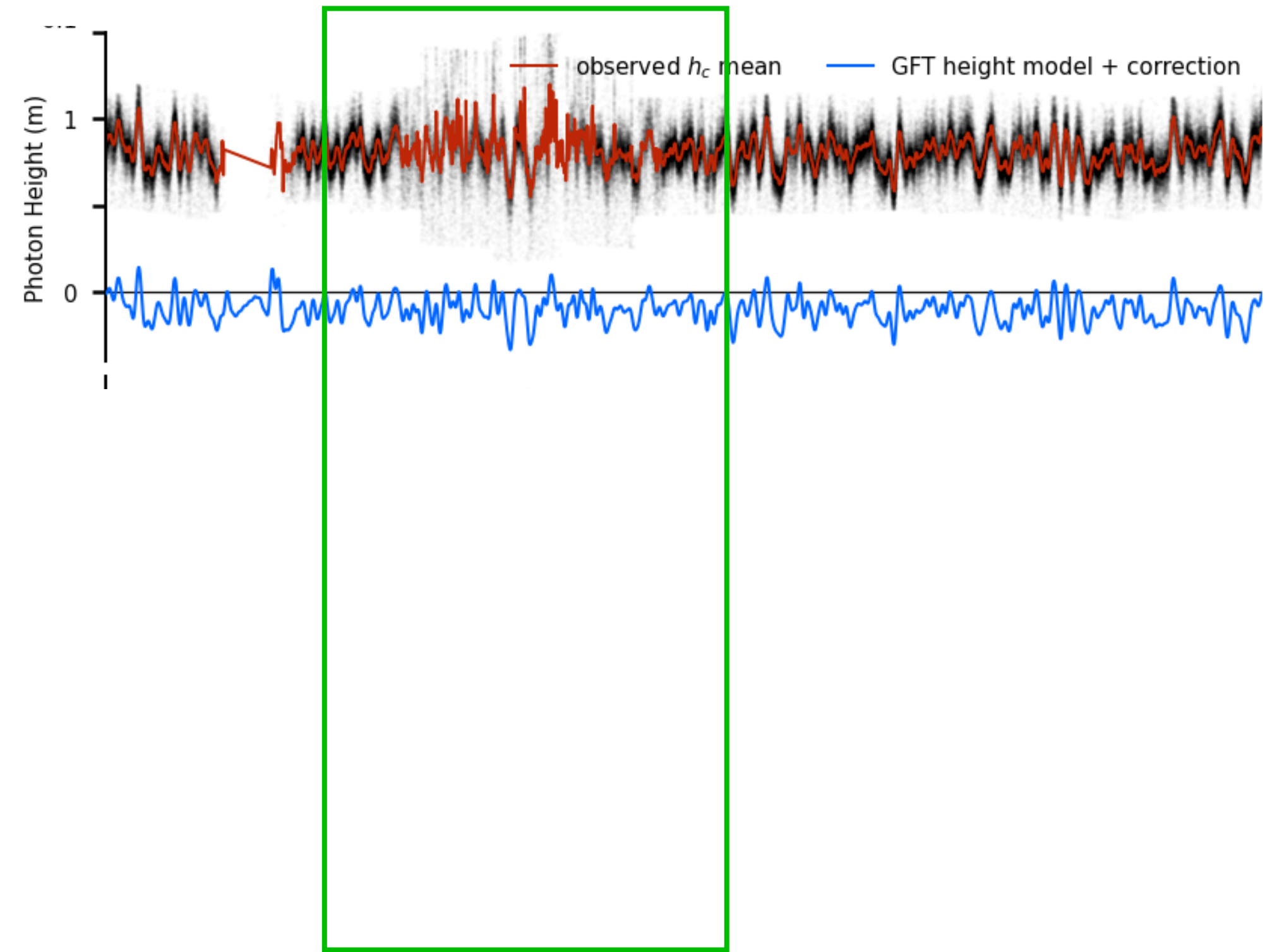


The Harmonic analysis along each beam show wave attenuation in sea ice



Each beam provides an estimate of the observed along-track wave spectra every 12.5 km

Filtered free board enable better sea ice products



Observed Photon heights

Sea ice height model

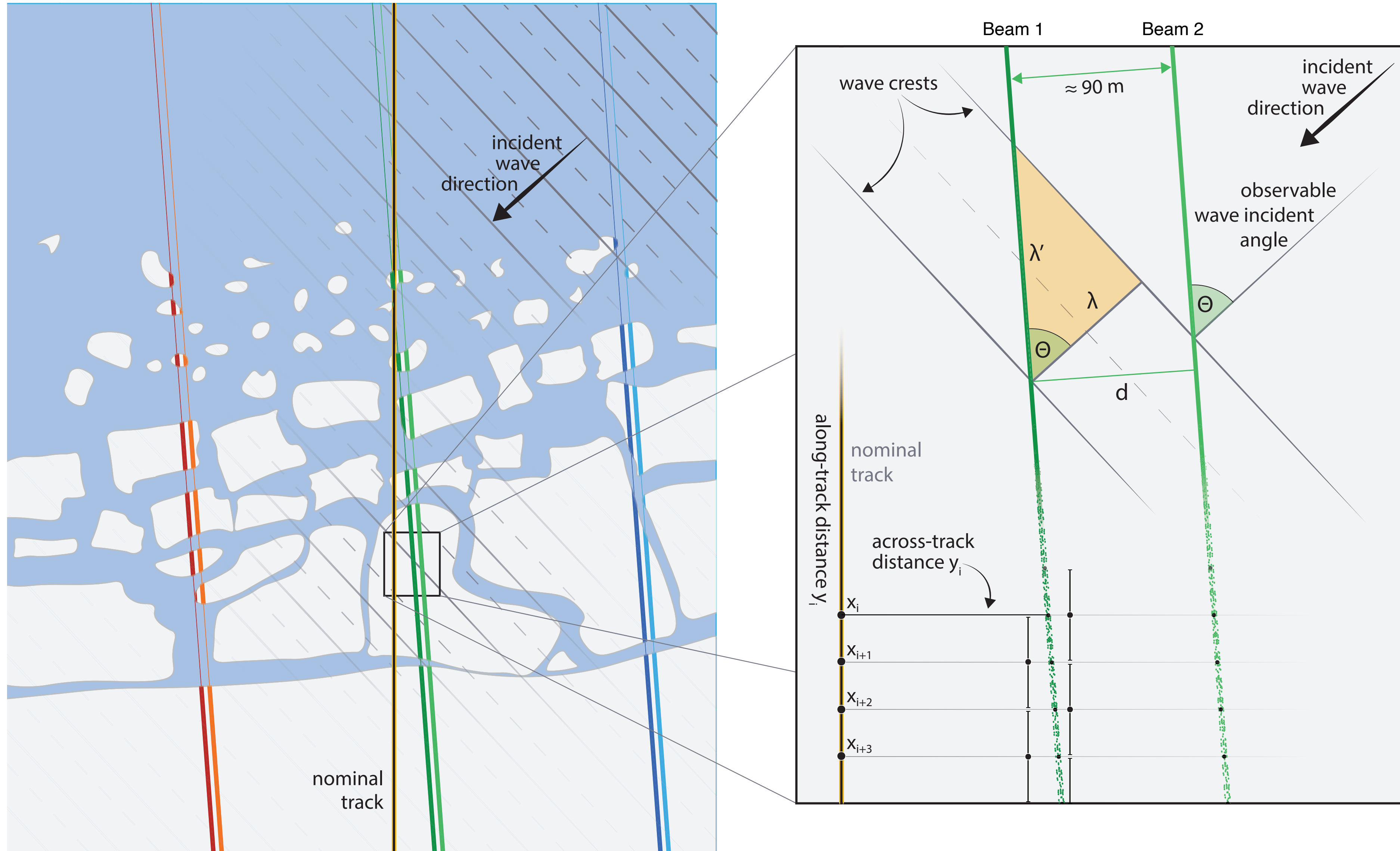
Residual surface heights

Along-track variance partitions

Surface waves take up to 50% of the photon height variance in the MIZ.

- 1. Surface wave energy and direction
For data-constrained wave attenuation parametrization
- 2. Sea Ice freeboard heights and flow sizes
For ice thickness and flow size estimates

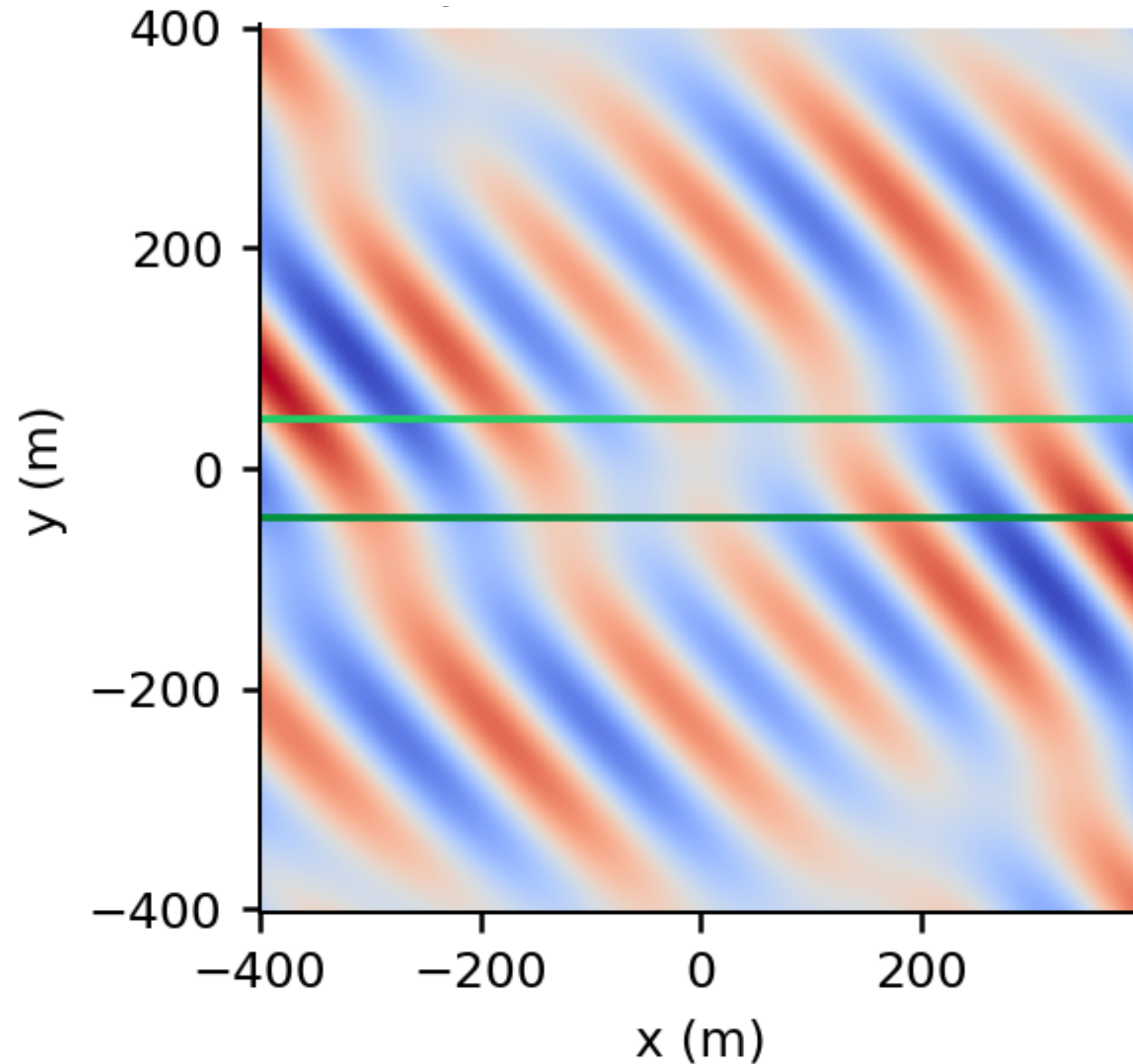
How to observe the wave direction from two laser beams?



- Very low signal-to-noise due to geometric and dynamic distortion effect!

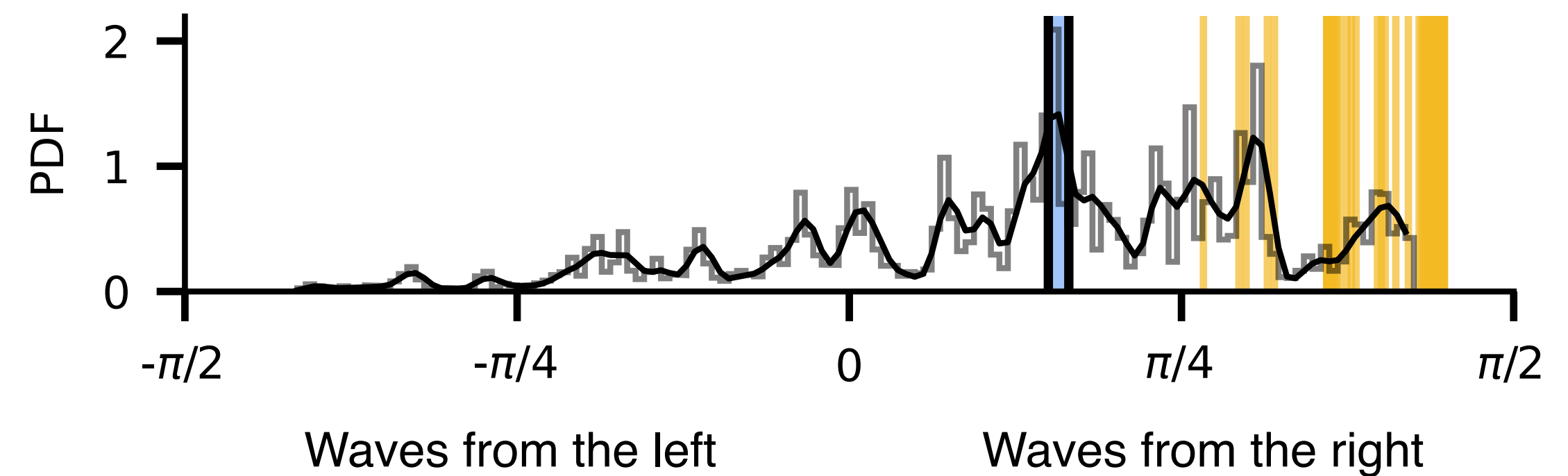
How to observe the wave direction from two laser beams?

Idealized waves under a beam pair



**Using non-linear optimization and
Metropolitan Hastings Sampling:**

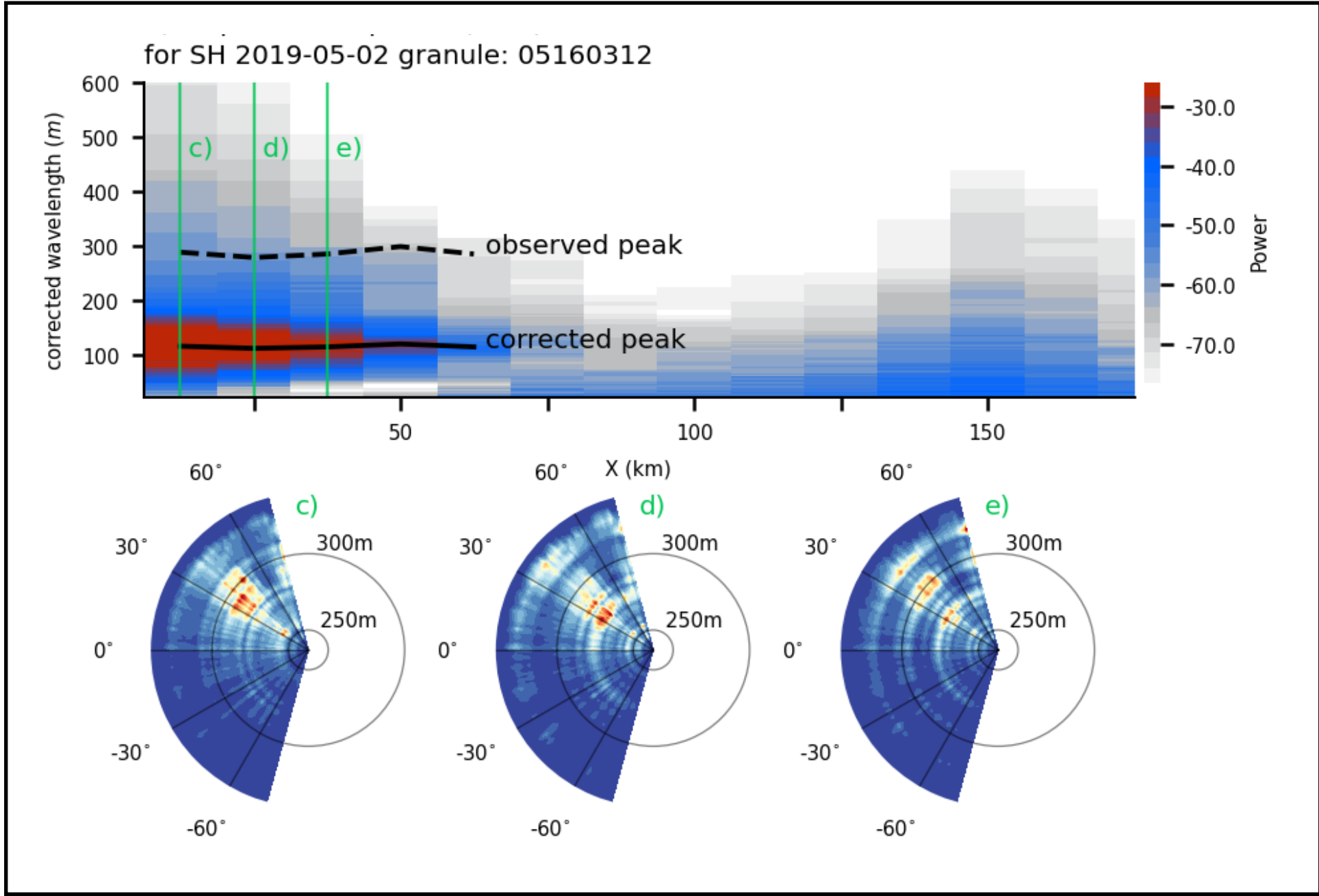
Maximum entropy of the wave incident angles



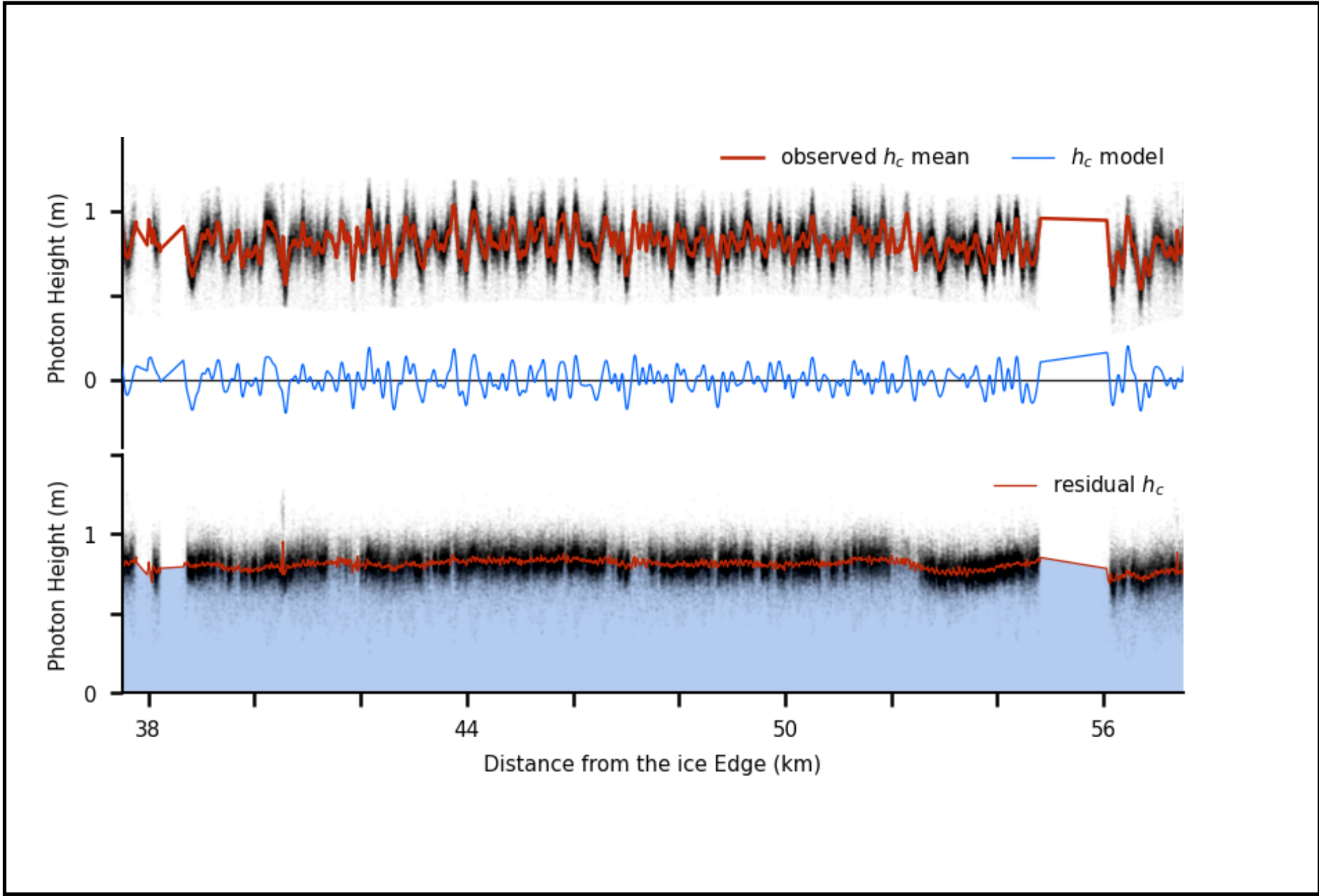
This turns ICESat-2 into a
directional wave observations

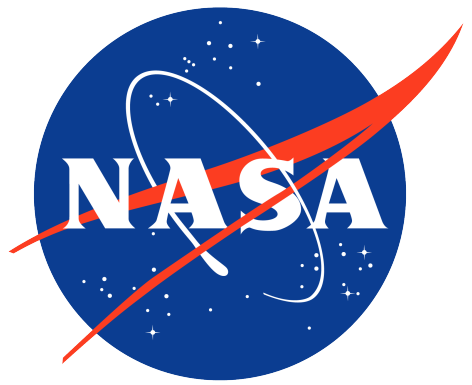
Directional wave spectra and residual heights

a) A directional wave buoy in sea ice every 12.5 km along the track.



b) Residual photon cloud for better sea ice products





Future work at WHOI & as NASA ICESat-2 Science team member

Many collaborations within WHOI

- Further development of inverse methods
- Validation with in-situ observations
- The impact of storms on sea ice

Data-driven parametrization of wave attenuation $\hat{B}(\omega, x) = \hat{B}_0 e^{-x} \alpha(\omega)$

Applying this algorithm as scale ($\sim 33 \times 10^3$ tracks) to create a database of wave-ice interaction

